PROGRAM CORE

Course Code	Course Title	L	T	P	S	C
22DS101017	**COMPUTER ARCHITECTURE AND ORGANIZATION**	3	-	-	-	3

Pre-Requisite 22CS101001 - Digital Logic Design

Anti-Requisite -

Co-Requisite -

COURSE DESCRIPTION: This course deals with basic function, structure and components of computer system, Central Processing Unit, Control Unit and I/O Unit. This course also imparts knowledge on different types of Memory Systems, Parallel Processing techniques and Multicore architectures with its issues.

COURSE OUTCOMES: After successful completion of the course, students will be able to:

CO1. To learn the fundamentals of computer organization and its relevance to classical and modern problems of computer design

CO2. To understand the structure and behaviour of various functional modules of a computer.

CO3. To learn the techniques that computers use to communicate with I/O devices.

CO4. To acquire the concept of pipelining and exploitation of processing speed.

CO5. To learn the basic characteristics of multiprocessors.

CO-PO-PSO Mapping Table:

Course Outcomes	Program Outcomes												Program Specific Outcomes			
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	-	-	-	-	-	-	-	-	-	-	-	3	-	-	-
CO2	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-	-
CO3	3	2	1	-	-	-	-	-	-	-	-	-	3	-	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-	-
CO5	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-	-
Course Correlation Mapping	**3**	**2**	**1**	-	-	-	-	-	-	-	-	-	**3**	-	-	-

Correlation Levels: 3: High; 2: Medium; 1: Low

COURSE CONTENT

Module 1 INTRODUCTION TO COMPUTER SYSTEM ***(09 Periods)***

Basic Structure of Computer: Computer Types, Functional Units, Basic operational Concepts, Bus Structure, Software, Performance, Multiprocessors and Multicomputer.

Machine Instructions and Programs: Numbers, Arithmetic Operations and Programs, Instructions and Instruction Sequencing, Addressing Modes, Basic Input/output Operations, Stacks and Queues, Subroutines, Additional Instructions.

Module 2 BASIC PROCESSING UNIT & ALU ***(09 Periods)***

Basic Processing Unit: Fundamental Concepts, Execution of a Complete Instruction, Multiple Bus Organization, Hardwired Control, Multi programmed Control, Micro programmed control unit.

Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Positive Numbers, Signed-operand Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations.

Module 3 MEMORY SYSTEMS ***(09 Periods)***

Basic Concepts, Semiconductor RAM Memories, Read-Only Memories, Speed, Size and Cost, Cache Memories, Performance Considerations, Virtual Memories, Memory Management, Requirements, Secondary Storage.

Module 4 INPUT/OUTPUT ORGANIZATION ***(09 Periods)***

Accessing I/O Devices, Interrupts, Processor Examples, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces. Hashing: Introduction to Hash Table, Static Hashing, Dynamic Hashing.

Module 5 PARALLEL ORGANIZATION ***(09 Periods)***

Parallel Processing: Basic Concepts, Data Hazards, Instruction Hazards, Influence on Instruction Sets. Large Computer Systems: Forms of Parallel Processing, Array Processors, The Structure of General-Purpose multiprocessors, Interconnection Networks.

Multicore Computers: Hardware performance issues, Software performance issues, Multicore organization, Intel Core i7-990X.

Total Periods: 45

EXPERIENTIAL LEARNING

1. Write a report on Product Specifications such as CPU, Memory, Processor Graphics and advanced technologies of Intel Core i9-12900K -New performance hybrid architecture. The World's Best Gaming Processor: Available with up to 16 cores and 24 threads, the new 12th Gen Intel Core processor family includes the world's best gaming processor, the Core i9-12900K, unleashing gaming experiences across top titles. (https://ark.intel.com/content/www/us/en/ark/products/134599/intel-core-i912900k-processor-30m-cache-up-to-5-20-ghz.html)
2. Write a report on OpenCL(Open Computing Language) framework used for writing programs that execute across heterogeneous platforms consisting of central processing units, graphics processing units, digital signal processors, field-programmable gate arrays and other processors or hardware accelerators.(https://www.khronos.org/api/opencl)

RESOURCES

TEXT BOOKS: